

UNIT 1: INTRODUCTION TO COMPUTER ORGANIZATION AND ARCHITECTURE

Introduction: Functional units of digital system and their interconnections, buses, bus architecture, types of buses and bus arbitration. Register, bus and memory transfer. Processor organization, general registers organization, stack organization and addressing modes.

◇ 1. Introduction

Computer Organization and Architecture (COA) deals with **how a computer system works internally** — how data moves, how instructions are executed, and how hardware components communicate.

Architecture = What the system *does* (design, capabilities).

Organization = How it *does it* (implementation details).

◇ 2. Functional Units of a Digital Computer

A **digital computer** performs basic operations like input, processing, storage, and output.

It consists of **five main functional units**:

Functional Unit	Description	Example
1. Input Unit	Takes data and instructions from user and converts them into machine-readable form.	Keyboard, Mouse
2. Output Unit	Converts processed data from machine form to human-readable form.	Monitor, Printer
3. Memory Unit	Stores data and instructions temporarily or permanently.	RAM, ROM
4. Arithmetic and Logic Unit (ALU)	Performs arithmetic (add, sub, etc.) and logic (AND, OR, NOT, etc.) operations.	Calculator in CPU
5. Control Unit (CU)	Controls and coordinates all operations of the computer.	CPU control circuitry

👉 Together ALU + CU = CPU (Central Processing Unit)

◇ 3. Interconnection of Functional Units

All these units communicate using **system buses**.

The **Control Unit** sends signals to coordinate activities.

The **Memory Unit** stores data/instructions.

The **ALU** processes them.

Input/Output Units exchange data with the external world.

◆ Example:

When you press a key on the keyboard → Input Unit sends data → CPU processes it → Output Unit displays it on screen.

◇ 4. Buses and Bus Architecture

◇ What is a Bus?

A **Bus** is a group of **parallel wires or lines** used to transfer data, addresses, or control signals between different components.

It acts like a **highway for data movement** inside the computer.

◇ Types of Buses

Type of Bus	Description	Example
1. Data Bus	Carries actual data between CPU, memory, and I/O devices.	8-bit, 16-bit, 32-bit, or 64-bit
2. Address Bus	Carries the memory address of data (to specify where to read/write).	Unidirectional
3. Control Bus	Carries control signals like Read/Write, Clock, Interrupts, etc.	Bidirectional

■ Analogy:

Imagine a delivery system:

Address Bus = house address

Data Bus = parcel (data)

Control Bus = delivery instructions (“deliver”, “pickup”)

◇ Bus Architecture Types

Type	Description	Example
1. Single Bus	All components share a common bus. Simple but slower.	Used in small systems

2. Multiple Bus	Separate buses for CPU–Memory and CPU–I/O, faster performance.	Used in modern PCs
3. System Bus	Combination of all (data, address, control).	Used in microprocessors

◇ 5. Bus Arbitration

When multiple devices want to use the **same bus** at the same time — we need **Bus Arbitration** to decide who gets control.

◇ Methods of Bus Arbitration

Method	Description	Example
1. Daisy Chaining (Serial Priority)	Devices are connected in a chain. Priority decreases as you go down the chain.	Simple but slower for low-priority devices
2. Centralized Arbitration	A central bus controller decides which device gets control.	Used in modern systems
3. Distributed Arbitration	Each device has its own control logic; they decide among themselves.	Complex but fast

◇ 6. Register, Bus, and Memory Transfer

◇ Registers

A **register** is a small, high-speed storage unit inside the CPU used to store temporary data or instructions.

Common CPU Registers:

Type	Purpose
AC (Accumulator)	Stores intermediate results
IR (Instruction Register)	Holds current instruction
PC (Program Counter)	Holds address of next instruction
MAR (Memory Address Register)	Holds address for memory access
MDR (Memory Data Register)	Holds data fetched from or to be written to memory
General Purpose Registers	Used for temporary data storage

◇ Register Transfer

Register Transfer means moving data from one register to another.

Represented symbolically as:

👉 R2 ← R1 (means contents of R1 are copied to R2)

◇ Bus and Memory Transfer

CPU uses **buses** to transfer data between **registers and memory**.

Example:

MAR ← PC → load address from Program Counter to Memory Address Register
MDR ← Memory[MAR] → read data from memory into Memory Data Register

◇ 7. Processor Organization

Processor = **ALU + CU + Registers + Interconnection Bus**

◇ Types:

Single Accumulator Organization:

One accumulator for all operations.

Example: ADD X → adds value at X to accumulator.

Simple but limited.

General Register Organization:

Multiple registers used for storing data.

Example: R0, R1, R2...

Faster data access and flexible.

Stack Organization:

Uses **Stack** (LIFO – Last In First Out).

Instructions like PUSH, POP.

Used in recursive functions and expression evaluation.

◇ 8. Addressing Modes

Addressing Mode = the way an instruction specifies the **location of data** (operand).

Mode	Description	Example	Meaning
1. Immediate	Operand is given directly in the instruction.	MOV R1, #5	R1 = 5
2. Direct	Address of operand is given.	MOV R1, 5000	R1 = value at memory[5000]
3. Indirect	Address of operand is stored in a register.	MOV R1, (R2)	R1 = value at memory[address in R2]

4. Register	Operand is in a register.	ADD R1, R2	$R1 = R1 + R2$
5. Indexed	Effective address = Base Register + Index.	MOV R1, 100(R2)	$R1 = \text{value at memory}[100 + R2]$

Summary of Unit 1

Topic	Key Points
Functional Units	Input, Output, Memory, ALU, CU
Bus Types	Data, Address, Control
Bus Arbitration	Daisy Chain, Centralized, Distributed
Registers	AC, PC, IR, MAR, MDR
Processor Org.	Single Accumulator, General Register, Stack
Addressing Modes	Immediate, Direct, Indirect, Register, Indexed

1. Computer Architecture vs Computer Organization

Basis	Computer Architecture	Computer Organization
Meaning	Describes <i>what</i> a computer does — the design and features visible to a programmer.	Describes <i>how</i> the system is built — internal working and implementation.
Focus	Logic, instruction set, addressing modes, data formats.	Hardware details like control signals, memory, buses, and circuits.
Concerned With	Structure and behavior of CPU as seen by programmer.	How architecture is implemented physically.
Example	Type of instruction set (RISC, CISC).	How control unit executes those instructions.
User Visibility	Visible to programmer.	Hidden from programmer.

2. Functional Units of a Computer

Unit	Function	Example
Input Unit	Takes data and instructions.	Keyboard, Mouse
Output Unit	Displays processed results.	Monitor, Printer
Memory Unit	Stores data temporarily/permanently.	RAM, ROM
ALU	Performs arithmetic & logic operations.	Addition, Comparison

Control Unit	Directs operations and timing.	Control signals
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(You can use this table for “Explain functional units” type questions.)

3. Types of Buses

Type	Direction	Purpose	Example
Data Bus	Bidirectional	Carries actual data.	8/16/32/64-bit data lines
Address Bus	Unidirectional	Carries address of data/instruction.	16-bit address bus can access 2^{16} locations
Control Bus	Bidirectional	Carries control signals (Read/Write/Interrupt).	Read, Write, Clock signals

4. Single Bus vs Multiple Bus Architecture

Basis	Single Bus	Multiple Bus
Number of Buses	One common bus for all components.	Separate buses for CPU–Memory and CPU–I/O.
Speed	Slower (data traffic congestion).	Faster (parallel transfers possible).
Cost	Cheaper, simple design.	Costlier, complex design.
Example Use	Small microcontrollers.	Modern PCs and processors.

5. Bus Arbitration Methods

Basis	Daisy Chaining	Centralized	Distributed
Control	Priority fixed serially (chain).	One central controller.	All devices have arbitration logic.
Priority	Fixed, highest priority near CPU.	Controlled by central arbiter.	Dynamic (decided collectively).
Speed	Slow for low-priority devices.	Moderate.	Fastest.
Complexity	Simple.	Moderate.	Complex.

6. Types of CPU Registers

Register	Purpose
PC (Program Counter)	Holds address of next instruction.
IR (Instruction Register)	Holds current instruction.
MAR (Memory Address Register)	Stores memory address to access.
MDR (Memory Data Register)	Temporarily holds data from/to memory.
AC (Accumulator)	Holds intermediate arithmetic/logic results.

General Purpose Registers	Temporary data storage.
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? 7. Processor Organization Types

Type	Features	Advantages	Example
Single Accumulator	One accumulator for all operations.	Simple hardware.	Early microprocessors.
General Register	Multiple registers for data storage.	Fast & flexible.	Intel x86, ARM.
Stack Organization	Uses stack (LIFO).	Simple instruction set.	Used in expression evaluation.

8. Addressing Modes

Mode	Example	Meaning	Use Case
Immediate	MOV R1, #5	Data = 5 directly in instruction.	Loading constants.
Direct	MOV R1, 5000	Operand at memory[5000].	Access fixed memory.
Indirect	MOV R1, (R2)	Operand at address in R2.	Pointer-based access.
Register	ADD R1, R2	Operands in registers.	Fastest access.
Indexed	MOV R1, 100(R2)	Effective address = 100 + R2.	Access array elements.



9. Memory vs Register

Basis	Memory	Register
Location	Outside CPU	Inside CPU
Speed	Slower	Very fast
Size	Large	Small (few bytes)
Cost	Cheap per bit	Costly per bit
Usage	Store programs and data	Temporary results during processing

10. Register Transfer vs Memory Transfer

Basis	Register Transfer	Memory Transfer
Between	Two registers	Between register and memory
Speed	Very fast	Slower

Control Signals	Internal CPU signals	Requires Read/Write signals
Example	R2 ← R1	MDR ← Memory[MAR]